

PIZON KHAN

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EDUCATION

Georgia Institute of Technology — M.S., Computer Science

GPA: 3.7 / 4.0

In progress. Concentration in Big Data Systems, Cloud Computing, Machine Learning, and Large Language Models.

Queens College, City University of New York — B.S., Computer Science & Applied Mathematics (Double Major) 2020

PROFESSIONAL EXPERIENCE

Webster Bank

New York City, NY

Director, Credit Analytics

Jul 2025 – Present

- Architected and own the Credit Data Mart, the bank's system of record for credit reporting — conformed loan-, customer-, and collateral-level tables fed by 15+ Snowflake pipelines producing daily and monthly point-in-time snapshots, consumed by 10+ enterprise teams (~100 users) and published each month-end as the basis for earnings and shareholder reporting.
- Built automated data-quality and anomaly-detection pipelines that surface outliers, inconsistencies, and reconciliation breaks before publication, cutting month-end remediation from 1–2 weeks to under 1 week and redirecting analyst capacity from error hunting to variance analysis.
- Designed and shipped the Credit Deal Tracker, a Snowflake Streamlit application processing ~60 executive credit deals per month: OCR and regex-based text and table extraction from deal memos, Cortex LLM summarization, and lifecycle tracking of approvals, comments, and change logs in a governed database.
- Extended the platform with an executive dashboard for month-over-month trend review, auto-generated committee decks, and an embedded LLM chatbot answering natural-language questions and generating charts over deal data — replacing a manual single-owner Excel and PowerPoint workflow, cutting per-deal effort from ~60 minutes to 5–10 and distributing intake across ~24 users.
- Lead infrastructure, tooling, and AI architecture decisions for the Credit organization and drive enterprise programs end to end — scoping and presenting to executive stakeholders, IT, and EPO, then serving as hands-on technical implementer.

Manager, Data Science

Feb 2024 – Jul 2025

- Led a team of 8 building in-house PD and LGD dual-risk-rating models for Commercial Real Estate, Sponsor & Specialty, and Asset-Based Lending portfolios, replacing licensed Moody's models and saving \$1.5M annually; binned-variable logistic regression with weights of evidence achieved ~89% (PD) and ~70% (LGD) ROC on CRE.
- Built a Python application that parses rich text and images to detect bias language in credit appraisal documents, cutting review time from ~60 minutes to under 2 minutes per document for a 15-person team and eliminating a ~\$25K/year third-party vendor subscription while improving detection accuracy.
- Developed a React and TypeScript loan-origination web application integrated with Snowflake for live data flow, operationalizing the dual-risk-rating models for commercial lending.

Data Scientist

Jun 2022 – Feb 2024

- Extracted and consolidated historical credit approval data across internal systems and third-party sources (TREPP), imputing gaps via k-nearest-neighbor clustering, to establish the analytical foundation for the bank's dual-risk-rating models.
- Partnered with credit underwriters to translate underwriting judgment into quantifiable risk drivers aligned to observed default and loss patterns across CRE, S&S, and ABL portfolios.

Whiterock.ai

Manhattan, NY

Data Scientist

Jan 2022 – May 2022

- Automated ingestion of Black Knight and third-party real-estate data using Apache Airflow on Google Cloud Platform, reducing processing time 20% and improving data quality; performed EDA and ETL on large property datasets to engineer risk drivers for home-price models.

ProMarketingHub

Queens, NY

Junior Data Scientist

Jul 2020 – Oct 2021

- Built ETL processes and k-means customer segmentation models in Python to power targeted marketing campaigns.

TECHNICAL SKILLS

Languages: Python, SQL, SAS, R, Java, C++, TypeScript, JavaScript, Bash, LaTeX

Data & Cloud: Snowflake (Snowpark, Streamlit, Cortex, Snowpipe, Tasks & Streams), AWS (SageMaker, S3, Lambda), GCP (BigQuery), Databricks, Apache Airflow, Spark, dbt

Data Engineering: ETL/ELT pipeline design, dimensional & data-mart modeling, point-in-time snapshots, slowly changing dimensions, orchestration, data quality and reconciliation frameworks, CI/CD, Git

AI & Machine Learning: Large Language Models, RAG, prompt engineering, NLP, OCR (Tesseract, ABBYY), scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, logistic regression, weights of evidence, clustering, model back-testing

Applications & Visualization: Streamlit, React, Pandas, NumPy, Tableau, Power BI, REST APIs, Docker, A/B testing